AIRFOIL SELECTION

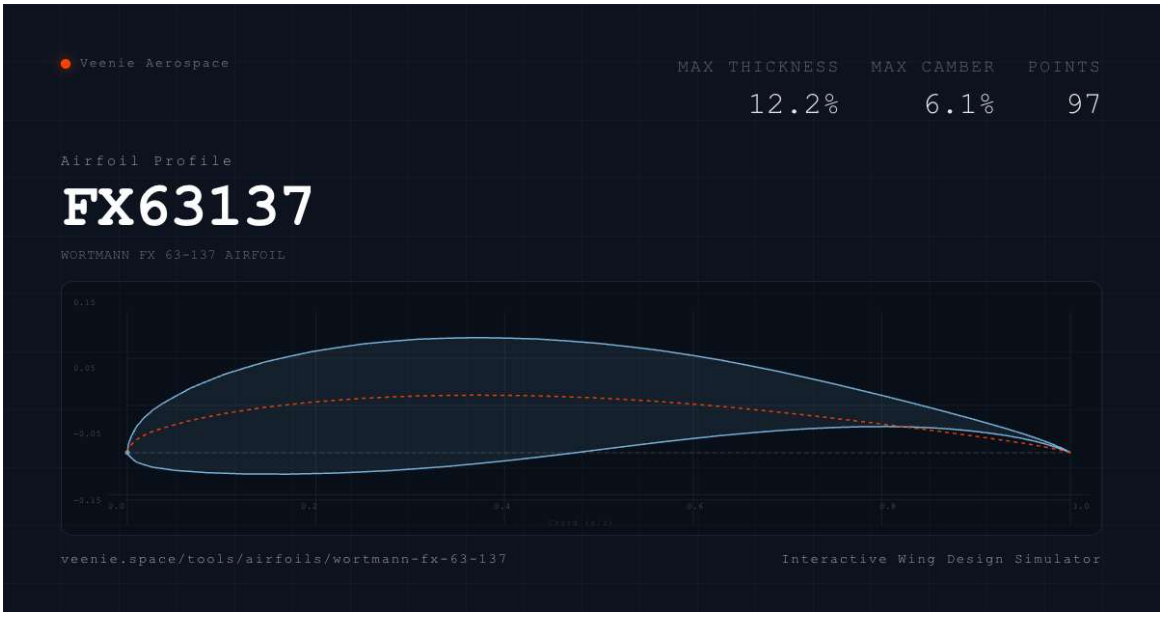
PROBLEM 1

Airfoil optimized for maximum payload while maintain stable flight:

Top 3 airfoil that can be used are

- Selig s1223
- Eppler E423
- Wortmann FX63-137

Wortmann FX-63-137 is the best for general use because it balances high lift (C=1.7) with a moderate pitching moment , a remarkably soft and progressive stall and a high lift to drag ratio across a wide range.



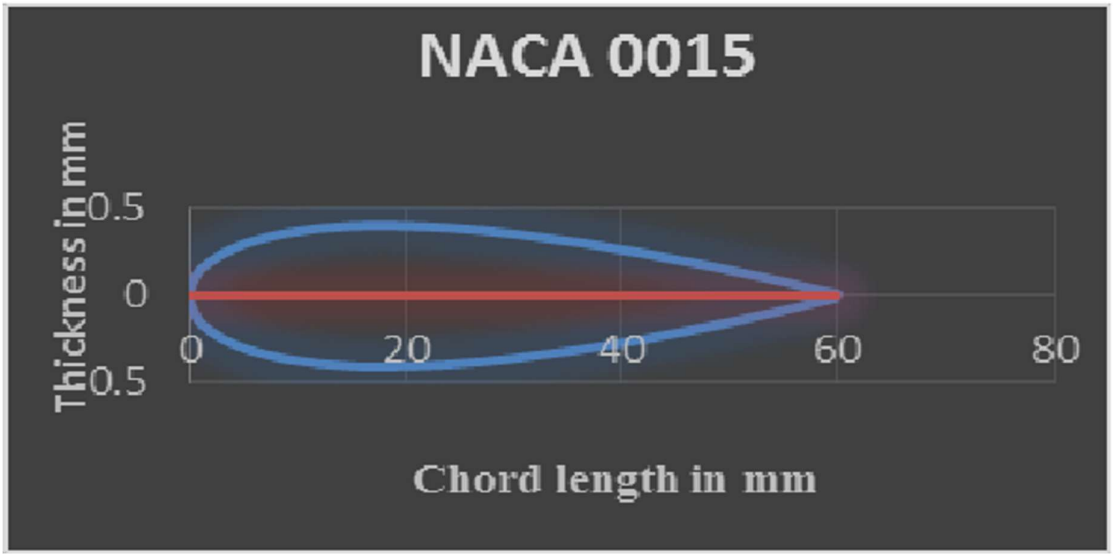
PROBLEM 2

Airfoil optimized for maneuverability (turns, rolls ,loops):

Top 3 airfoil that can be used are :

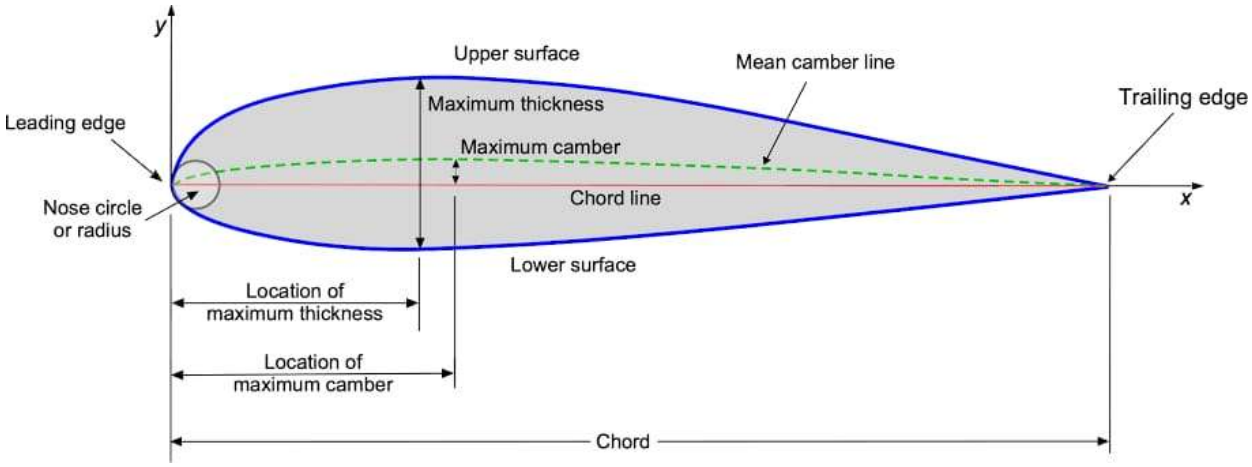
- NACA 0012
- Eppler 374
- NACA 0015

The NACA 0015 is the best choice for maximum aerobatic maneuver ability and high-g authority. Maneuver ability in aircraft design depends primarily on high maximum lift coefficient for tight turn radii and high-g pulls, a wide usable angle of attack range, symmetrical behaviour (identical upright and inverted flight characteristics), and a forgiving stall



	Problem 1	Problem 2
Lift coefficient	Max lift coefficient	Not specified
Drag coefficient	Less required	High requirement
Lift to drag ratio	High ratio would be preferable	Ratio should be less
camber	Positive, high and forward to mid camber	Low camber, ideally zero camber(symmetric)
Thickness to chord ratio	15-18%	5-12%

AIRFOIL GEOMETRY:



An airfoil geometry is defined by its front leading edge, sharp rear trailing edge, and the straight chord line connecting them. Its curvature is mapped by the mean camber line running midway between surfaces, with camber measuring the distance from that centre line to the chord, while thickness is the perpendicular distance between the upper and lower surfaces, and the leading-edge radius dictates nose roundness.

Can single airfoil staisfies both requirement ?

A single fixed airfoil cannot maximize both because their geometric demands are directly contradictory. Carrying maximum payload demands an aggressively curved, high-camber airfoil to generate high low-speed lift, which intrinsically creates a heavy nose-down pitching moment and completely destroys inverted flight capability. Conversely, maximum maneuverability demands a completely symmetrical airfoil with zero pitching moment so the aircraft responds

identically upright and inverted without pitch trim wander. Choosing high camber makes agile aerobatic rolls and inverted loops impossible, while choosing a symmetrical shape drastically cuts the maximum lift coefficient, sacrificing the heavy lifting capacity.

AIRFOIL VS AIRCRAFT

Airfoil characteristics describe the 2D cross-sectional shape of the wing, defining unitless properties like camber, thickness, and profile lift/drag coefficients. In contrast, whole-aircraft factors represent the complete 3D flying machine with finite dimensions, mass, and hardware. While the airfoil sets the baseline aerodynamic efficiency and stall potential, whole-aircraft factors—such as wing loading (W/S), motor thrust , control surface sizing, and structural spars—determine whether the plane can physically support payload, withstand g-forces, accelerate, and maneuver in three-dimensional space.